Kaelin Wulf

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Education

Reed College **August 2020 – May 2024**
B.A., Neuroscience; Cumulative GPA: 3.51
Commendation for Excellence in Scholarship
Thesis: Glia Cell Dynamics in the Zebrafish Optic Tectum

Research Experience

Post-Baccalaureate Microscopy Engineer **July 2024 - present**
Allen Institute for Neural Dynamics; Advisor: Adam Glaser

- Developed, characterized, aligned and operated custom large-scale light-sheet microscopes for imaging whole expanded mouse brains.
- Authored an imaging protocol and detailed documentation wiki on GitHub; and drafted a manuscript on the microscope and properties of expansion microscopy (in-progress).
- Improved optical and mechanical designs through CAD modeling, machining, and prototyping novel optical assemblies.
- Lead the assembly of multiple microscope systems and independently performed alignment and troubleshooting.
- Presented work at O-STEM, Optica Bio-Photonics Congress, Lake Conference, and SPIE Photonics West (planned Jan 2026).

Senior Thesis Student **August 2023 - May 2024**
Reed College Neuroscience Department; Advisor: Kara Cerveny

- Conducted a yearlong independent thesis research project studying the cell dynamics and patterns of microglia and oligodendrocyte progenitor cells (OPCs) in the optic tectum of larval zebrafish.
- Optimized an image acquisition and analysis pipeline to work with large data on limited hardware and trained team on best imaging practices.
- Produced a ~100-page written thesis with oral defense; presented results as a poster at the SDB, 2024 Northwest Regional Meeting, and as a selected talk to the Reed College community.

Undergraduate Research Fellow **May - August 2023**
Reed College Neuroscience Department; Advisor: Kara Cerveny

- Led a summer research project on the topic of cell death and microglia cell dynamics in the optic tectum of larval zebrafish.
- Generated transgenic crosses and single-cell injectable plasmid vectors and imaged larvae using confocal and light-sheet microscopy.
- Presented findings as a poster at the Society for Neuroscience 2023 conference.

Undergraduate Research Fellow **May - August 2022**
Reed College Biology Department; Advisors: Erik Zornik & Charlotte Barkan

- Proposed, managed, and executed a collaborative research project investigating androgen-dependent neuroanatomy in *Xenopus laevis* vocal-motor circuits.
- Conducted animal surgeries, dissections, and histological preparations including IHC and tissue clearing for light-sheet imaging.
- Presented results with a poster at the Society for Neuroscience 2022 conference

Student Intern **June - August 2020**
Air Force Research Laboratory; Advisor: Wellesley Pereira

- Collected and analyzed photometric telescope images of satellites to develop an approach to extracting fundamental physical properties of space objects.
- Collaborated with remote teams under virtual limitations to design novel methods for multispectral data analysis.

Leadership Experience

Confidential Advocate and Coordinator **September 2022 – May 2024**
Reed College SHARE program

- Provided confidential advocacy and support for survivors of interpersonal violence.
- Facilitated weekly full team meetings, and led working groups focused on education and reshaping campus culture around topics of sexual violence.
- Organized community events and led workshops on consent, supporting survivors, and the neurobiology of trauma.
- Delivered workshops on survivor support, consent culture, and the neurobiology of trauma.

Honors/Awards

Summer Undergraduate Research Fellowship 2022 & 2023
The Beckman Foundation, and the Helen Stafford Summer Research Fellowship Fund

Undergraduate Research Opportunity Grant 2022 & 2023
The Galakatos Science Research Fund, and Paul K. Richter & Evalyn Elizabeth Cook Richter Memorial Fund

Initiative Grant in Undergraduate Research 2024
Reed College, Neuroscience Department

Microscopy Image Competition 2024
Northwest Developmental Biology Conference, Nikon Exhibition

Publications

Wulf, K., Jiang, X., Glaser, A., et al. "An optimized expansion-assisted selective plane illumination microscope for nanoscale imaging of centimeter scale tissues". (in preparation)

Conference Presentations and Proceedings

Wulf, K., Jiang, X., Woodard, M., Mwaniki, W., Glaser, A. "Optimized expansion-assisted selective plane illumination microscopy (ExA-SPIM)". *SPIE Photonics West*, 2026. Paper 13859-8. [Abstract & Talk]

Cao, K., Flannery, R., Javeri, R., **Wulf, K.,** Grotz, P., Arshadi, C., Ouellette, N., Hooper, M., Woodard, M., Jiang, X., Friedmann, D., Roy, L., Summers, M., Tasic, B., Svoboda, K., Glaser, A., Chandrashekar, J. "Organization of the thalamic reticular nucleus across spatial scales" *Society for Neuroscience, 2025.*

Wulf, K., Javeri, R., Vasquez, S., Arshadi, C., Ouellette, N., Jiang, X., Baka, J., Kovacs, G., Woodard, M., Seshamani, S., Cao, K., Clack, N., Grim, A., Turschak, E., Liddell, A., Rohde, J., Grotz, P., Logsdon, M., Feng, D., Svoboda, K., Chandrashekar, J., and Glaser, A. "New technologies for mapping centimeter scale tissues with nanoscale resolution" *Lake Conference* 2025, [Poster].

Cao, K., Flannery, R., Javeri, R., **Wulf, K.,** Grotz, P., Arshadi, C., Ouellette, N., Hooper, M., Woodard, M., Jiang, X., Friedmann, D., Roy, L., Summers, M., Tasic, B., Svoboda, K., Glaser, A., Chandrashekar, J. "Multiscale molecular anatomy of the thalamic reticular nucleus" *Lake Conference* 2025.

Wulf, K., Jiang, X., and Glaser, A. "Characterization of a new expansion-assisted selective plane illumination microscope". *Optica Biophotonics Congress* 2025, Technical Digest Series (Optica Publishing Group, 2025), paper DM3A.2. [Abstract & Talk]

Jiang, X., **Wulf, K.**, Woodard, M., Mwaniki, W., Glaser, A. "A new expansion-assisted selective plane illumination microscope for nanoscale imaging of centimeter-scale tissues". *Optica Biophotonics Congress* 2025, Technical Digest Series (Optica Publishing Group, 2025), paper NM5C.2.

Wulf, K., Jiang, X., and Glaser, A. "Expansion-assisted selective plane illumination microscopy for nanoscale imaging of centimeter-scale tissues". *O-STEM*, 2024. [Abstract & Poster]

Wulf K. and Cerveny K. "Effects of innervation on oligodendrocyte and microglia cell dynamics in the zebrafish visual system". *Society for Developmental Biology, Northwest Regional Meeting*, 2024. [Poster]

Wulf K., Yue J., and Cerveny K. "Dynamics of cell death in the denervated zebrafish optic tectum". Faculty for Undergraduate Neuroscience; *Society for Neuroscience*, 2023. [Poster]

Wulf, K., Truong, B.B., Zornik E., and Barkan C. "Examining Androgen Induced Changes to Vocal Neurons of Female Xenopus Laevis". Faculty for Undergraduate Neuroscience; *Society for Neuroscience,* 2022. [Poster]

Professional Associations

Society for Neuroscience (SFN) 2022-present

Society of Photo-Optical Instrumentation Engineers (SPIE) 2025-present

Optica 2024-present

Skills

<u>Engineering</u>: Optical alignment, Basic CAD design, Soldering, Manual machining, Testing and characterizing optics, Specifying/purchasing Parts, Mechanical construction

<u>Wet Lab</u>: Animal surgery/dissection, PCR, Genotyping, Plasmid cloning, Microinjections, Tissue sectioning, Immunolabeling, Tissue clearing

<u>Software</u>: Python, R, Photoshop/Illustrator, 3D Image Analysis (Imaris), ImageJ/FIJI